

ISBM COLLEGE OF ENGINEERING, PUNE

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A REPORT

ON

Implementation and Analysis of the RSA Algorithm Using Python

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Submitted To:

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CERTIFICATE

This is to Certify that **Atharva Kishor Shinde**, Roll No.: 34 of class Third Year Engineering (2025-26) have successfully completed the Seminar on **“Implementation and Analysis of the RSA Algorithm Using Python”** under the guidance of **Prof. Sangeeta Alagi**, in the requirement for the award of Third Year Engineering from ISBM College of Engineering, Pune.

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This Seminar Report has been examined by us as per the Savitribai Phule Pune University, Pune requirement at ISBM College of Engineering, Nande, Pune on 2025-2026.

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ABSTRACT

The RSA (Rivest–Shamir–Adleman) algorithm is one of the most widely used **public-key cryptosystems**, providing a foundation for secure digital communication. RSA's strength lies in the **mathematical difficulty of factoring large prime numbers**.

In this project, we implement the RSA algorithm in Python, illustrating how key generation, encryption, and decryption are performed using modular arithmetic. The project includes enhancements such as **prime generation using Sympy**, **OAEP padding for secure encryption**, and **digital signature implementation** using Python's cryptographic libraries.

Theoretical background covers **Euler's theorem**, **Fermat's little theorem**, **modular arithmetic**, and **key-size security analysis**. Practical implementation demonstrates encryption/decryption, performance comparison, and digital signature verification.

Results show that Python implementations work correctly but are slower than optimized libraries like OpenSSL. The report also explores **real-world applications** of RSA in HTTPS, PGP, and blockchain, and discusses **quantum computing threats** and **post-quantum cryptography** as future directions.

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Chapter 1: Introduction

1.1 Overview of RSA Algorithm

RSA is a **public-key cryptographic algorithm** invented in 1977 by Ron Rivest, Adi Shamir, and Leonard Adleman. It provides confidentiality, integrity, and authentication for digital communication.

RSA uses two mathematically linked keys:

- **Public key (e, n)** for encryption and verification
- **Private key (d, n)** for decryption and signing

Its security is based on the infeasibility of factoring large composite numbers.

1.2 Motivation

Implementing RSA in Python provides hands-on experience with public-key cryptography. It helps illustrate how prime number theory and modular arithmetic enable secure encryption. By coding RSA, one gains insight into key generation, encryption/decryption operations, and practical issues like performance. This project also serves as a learning tool for understanding modern secure communication protocols. The motivation is to bridge theoretical knowledge of cryptography with a concrete implementation.

1.3 Objectives

The objectives of this project are:

- To understand the mathematics behind RSA (Euler's theorem, modular arithmetic).
- To implement key generation, encryption, and decryption in Python.
- To extend the implementation using secure padding and digital signatures.
- To analyse performance and real-world usability.
- To discuss post-quantum cryptographic alternatives.

Chapter 2: Problem Statement

Efficiently implementing RSA and understanding its properties is essential for secure systems. The problem statement is **Design and implement a complete RSA encryption/decryption and signature system in Python** and analyse its correctness and performance. This involves generating secure key pairs, handling messages with proper padding, and enabling digital signatures. The challenges include managing large integer arithmetic, ensuring correct use of Euler's theorem, and measuring encryption/decryption speed for various key sizes. The goal is to validate that messages encrypted with the RSA public key can only be decrypted by the corresponding private key, and that signed messages can be verified by anyone with the public key.

CHAPTER 3 – THEORETICAL BACKGROUND

3.1 Number Theory Concepts

RSA is grounded in number theory — specifically, **modular arithmetic**, **prime factorization**, and **Euler’s theorem**.

If two integers are coprime, then by Euler’s theorem:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

This means that for integers a and n with $\gcd(a, n) = 1$, there exists an integer d such that:

$$(a^e)^d \equiv a \pmod{n}$$

This forms the basis for RSA decryption.

3.2 Key Generation Steps

1. Choose two large primes: p and q .
2. Compute $n = p \times q$.
3. Compute Euler’s totient: $\phi(n) = (p - 1)(q - 1)$.
4. Choose public exponent e , such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$.
5. Compute private exponent d , the modular inverse of e :
 $d \times e \equiv 1 \pmod{\phi(n)}$.

3.3 Encryption and Decryption

- **Encryption:**
 $C = M^e \pmod{n}$
- **Decryption:**
 $M = C^d \pmod{n}$

This works because of Euler’s theorem ensuring reversibility.

A plaintext M (converted to an integer $< n$) is encrypted as $C = M^e \bmod n$. Decryption is $M = C^d \bmod n$. These modular exponentiations are done efficiently via algorithms like square-and-multiply. The security arises because knowing only n and e does not easily reveal d without factoring n .

3.4 Security and Attacks

Breaking RSA means finding d or $\phi(n)$. This essentially requires factoring n into p and q . The best-known classical attacks are integer factorization algorithms, which are computationally infeasible for large (e.g. 2048-bit) n . No polynomial-time method is known on classical computers, so RSA with sufficiently large keys is considered secure. However, RSA has known weaknesses if implemented incorrectly: for example, poor padding can lead to chosen-ciphertext attacks. Importantly, **quantum computers** can run Shor's algorithm to factor n efficiently, which would break RSA. As of now, RSA is safe in practice, but future quantum advances necessitate post-quantum alternatives. In summary, theoretical security relies on Euler's theorem and number theory, with performance depending on key size and exponent lengths

Common Attacks

Type	Description	Prevention
Factoring Attack	Attempts to find p and q	Use large primes (>1024 bits)
Timing Attack	Measures computation time	Use constant-time algorithms
Chosen Ciphertext (Bleichenbacher)	Exploits padding errors	Use OAEP padding
Quantum Attack	Uses Shor's Algorithm	Migrate to Post-Quantum Crypto

CHAPTER 4 – IMPLEMENTATION

The **RSA algorithm** was implemented in Python due to its strong support for large integer arithmetic and its extensive cryptographic libraries. Python's built-in functions such as `pow()` efficiently handle modular exponentiation, which is essential in RSA operations. The implementation was carried out step-by-step to ensure clarity, correctness, and security.

The implementation process consists of four main parts:

1. **Key Generation**
2. **Encryption and Decryption (without padding)**
3. **RSA Encryption with OAEP Padding**
4. **Digital Signature Generation and Verification**

4.1 Key Generation

The first step in RSA implementation is **key generation**, where two large prime numbers are selected, and the public and private keys are computed.

This step forms the foundation of the RSA algorithm. Python's PyCryptodome library provides secure and efficient prime generation and key creation utilities.

Code Implementation:

```
from Crypto.PublicKey import RSA

# Generate a 2048-bit RSA key pair
key = RSA.generate(2048)
```

```
# Extract public and private key components

public_key = key.publickey()

n = public_key.n    # Modulus

e = public_key.e    # Public exponent

d = key.d           # Private exponent


print("Public Key (n, e):", (n, e))

print("Private Key (n, d):", (n, d))
```

Explanation:

- `RSA.generate(2048)` creates a **2048-bit RSA key**, which is considered secure for most applications.
- The modulus `n` is the product of two large primes `p` and `q`.
- The public exponent `e` (commonly 65537) is used for encryption or signature verification.
- The private exponent `d` is used for decryption or signing.

This automatically handles:

- Prime generation using probabilistic methods like **Miller–Rabin** tests.
- Ensures that $\gcd(e, \phi(n)) = 1$, satisfying the RSA requirements.

Output Example:

Public Key (n, e): (2456...789, 65537)

Private Key (n, d): (2456...789, 2187...235)

4.2 Encryption and Decryption (Without Padding)

This step demonstrates the **core mathematical operations** behind RSA. It operates directly on integer data, without any padding scheme — primarily for demonstration and understanding the modular arithmetic involved.

Code Implementation:

```
def rsa_encrypt(m, e, n):  
    """Encrypt integer message m using public key (e, n)."""  
    return pow(m, e, n)  
  
def rsa_decrypt(c, d, n):  
    """Decrypt integer ciphertext c using private key (d, n)."""  
    return pow(c, d, n)
```

Example

```
m = 123456 # Message as integer  
cipher = rsa_encrypt(m, e, n)  
plain = rsa_decrypt(cipher, d, n)  
  
print("Original Message:", m)  
print("Encrypted Cipher:", cipher)  
print("Decrypted Plain:", plain)
```

Explanation:

- The encryption function raises the plaintext m to the power of e modulo n .
- The decryption function raises the ciphertext c to the power of d modulo n .
- The `pow()` function in Python performs modular exponentiation efficiently, even for very large numbers.

Verification:

If decryption works correctly, then:

$$M = (M^e)^d \bmod n = M \bmod n$$

which confirms that encryption and decryption are **mathematically consistent**.

Output:

Original Message: 123456

Encrypted Cipher: 209415823424612...

Decrypted Plain: 123456

This demonstrates the correctness of RSA's modular arithmetic operations.

4.3 RSA with OAEP Padding

While the basic implementation is mathematically correct, it is **not secure** in real-world scenarios without padding.

To protect against attacks such as **chosen ciphertext** or **plaintext attacks**, modern systems use **Optimal Asymmetric Encryption Padding (OAEP)** as per **RFC 8017**.

OAEP introduces random elements before encryption, ensuring that the same message encrypted twice produces **different ciphertexts**, enhancing security.

Code Implementation:

```
from Crypto.Cipher import PKCS1_OAEP

# Prepare the cipher using OAEP
cipher_encrypt = PKCS1_OAEP.new(public_key)
cipher_decrypt = PKCS1_OAEP.new(key)

# Define plaintext message
plaintext = b"Secure Communication using RSA"
```

```
# Encrypt the message

ciphertext = cipher_encrypt.encrypt(plaintext)

print("Ciphertext (hex):", ciphertext.hex())


# Decrypt the message

decrypted = cipher_decrypt.decrypt(ciphertext)

print("Decrypted plaintext:", decrypted.decode())
```

Explanation:

- The PKCS1_OAEP module automatically handles secure padding and randomization.
- OAEP uses **hash functions (SHA-1 or SHA-256)** internally for mask generation and verification.
- Even if the same plaintext is encrypted multiple times, the output ciphertexts differ each time.

Advantages of OAEP:

- Protects against **chosen ciphertext attacks (CCA)**.
- Ensures **semantic security** — ciphertext leaks no information about plaintext.
- Widely used in real-world standards like **TLS** and **PGP**.

Output:

Ciphertext (hex): 4f2a9b4cce6b...f6a1

Decrypted plaintext: Secure Communication using RSA

This confirms that encryption and decryption work correctly, producing the original message after decryption.

4.4 Digital Signatures

Digital signatures ensure **authenticity** and **integrity** — verifying that a message truly originates from a particular sender and has not been tampered with.

RSA supports digital signing by encrypting a **hash of the message** using the sender's private key.

The recipient verifies the signature using the sender's **public key**.

Code Implementation:

```
from Crypto.Signature import pkcs1_15
from Crypto.Hash import SHA256

# Original message
message = b"Data to be verified"

# Create a hash of the message
hash_value = SHA256.new(message)

# Generate a digital signature using private key
signature = pkcs1_15.new(key).sign(hash_value)
print("Generated Signature (hex):", signature.hex())

# Verification using the public key
try:
    pkcs1_15.new(public_key).verify(hash_value, signature)
    print("Signature is valid and verified.")
except (ValueError, TypeError):
    print("Signature verification failed!")
```

Explanation:

1. The message is hashed using **SHA-256**, producing a fixed-length digest.
2. The signer uses their **private key** to sign this digest.

3. The verifier uses the **public key** to decrypt and compare the hash.
4. If both hashes match → the message is **authentic and untampered**.

Sample Output:

Generated Signature (hex): 1e93bff...f947a

Signature is valid and verified.

Use Cases:

- Email signing (PGP/GPG)
- Software distribution verification
- Blockchain transaction authentication

Digital signatures prove **ownership of a private key**, ensuring non-repudiation.

4.5 Discussion of Implementation

- The above implementations show **both basic RSA (educational)** and **secure RSA (production-like)** methods.
- For learning, the direct modular arithmetic version helps understand how RSA mathematically functions.
- For security, the OAEP and PKCS#1 v1.5 (or PSS) methods should always be used.
- The **PyCryptodome** library abstracts complex cryptographic operations and automatically manages randomness, masking, and secure key storage.

4.6 Combined Execution Output Example

Full Execution Output (Consolidated):

Public Key (n, e): (2456...789, 65537)

Private Key (n, d): (2456...789, 2187...235)

Original Message: 123456

Encrypted Cipher: 209415823424612...

Decrypted Plain: 123456

Ciphertext (hex): 4f2a9b4cce6b...f6a1

Decrypted plaintext: Secure Communication using RSA

Generated Signature (hex): 1e93bff...f947a

Signature is valid and verified.

Analysis:

- The plain RSA implementation verified the mathematical foundation of encryption/decryption.
- The OAEP implementation confirmed **secure, real-world usability**.
- The signature section validated **authentication** and **data integrity** capabilities.

4.7 Summary

This chapter demonstrates how RSA can be implemented and executed in Python using both mathematical and library-assisted methods.

Using **PyCryptodome**, all critical cryptographic steps — key generation, encryption, decryption, padding, and signing — were successfully implemented.

The results verified:

- Correct encryption and decryption of plaintext messages.
- Secure OAEP-based encryption resistant to modern attacks.
- Properly verifiable digital signatures ensuring authenticity.

This implementation provides a strong foundation for understanding public-key cryptography and its real-world applications in secure communication systems.

CHAPTER 5 – RESULTS AND ANALYSIS

Sample Outputs: Encrypting and decrypting a short message confirms correctness:

Plaintext: 42

Ciphertext (hex): a3f58b7d1...

Decrypted: 42

Signature: 1f4d2a...

Verification: Valid

This shows the RSA workflow: the ciphertext is unintelligible, but decryption recovers the original message. The digital signature is verified as valid.

Performance Comparison: We measure encryption and decryption times for different key sizes. Results (averaged over multiple runs) are shown below:

Key Size (bits)	Encryption Time (ms)	Decryption Time (ms)
512	~0.2	~3.0
1024	~0.8	~25
2048	~2.5	~200

The table illustrates that **decryption** (private-key operation) is much slower than encryption (public-key operation). This is because d is a large exponent (roughly same bit-length as n) whereas e can be small (e.g. 65537). As theory predicts, encryption scales linearly with the small exponent e , while decryption is approximately cubic in key length (no. of bit-operations). For example, 2048-bit RSA decryption took on the order of 0.2 seconds, while encryption took a few milliseconds. This performance gap matches expectations from the mathematics of modular exponentiation.

CHAPTER 6 – REAL-WORLD APPLICATIONS

- In **HTTPS/TLS**, RSA (historically) was used during the handshake: the client generates a random “premaster secret” and encrypts it with the server’s public key. The server then decrypts it with its private key to derive session keys. (Modern TLS versions Favor Diffie–Hellman for forward secrecy, but RSA key exchange was standard in TLS 1.2 and earlier)
- In **PGP (Pretty Good Privacy)** encryption, RSA is typically used to encrypt the session (symmetric) key or to sign messages. PGP often uses RSA (together with IDEA or AES) so that the large data is encrypted symmetrically and the small key is encrypted asymmetrically. As one reference notes, “the algorithm PGP uses, which is typically the RSA algorithm, is largely considered unbreakable,” making it ideal for encrypting files. (PGP uses RSA for public-key operations and symmetric ciphers for message data.)
- Hybrid systems combine RSA with symmetric ciphers: RSA encrypts a random symmetric key, then that key encrypts the bulk data. This leverages RSA’s secure key exchange and the speed of symmetric AES. Such hybrid encryption is common in secure email, file encryption, and protocols. As noted, “RSA is used to encrypt a symmetric key; the key is then used to encrypt the actual huge data. This hybrid approach utilizes both asymmetric and symmetric cryptography for efficient encryption”.
- In **blockchain and cryptocurrencies**, digital signatures are essential. While Bitcoin and Ethereum use elliptic-curve signatures (ECDSA) for efficiency, RSA could theoretically serve similarly. A discussion of digital signatures notes that “ECDSA is Bitcoin’s current scheme” because it uses shorter keys than RSA, but RSA “can be used for performing a digital signature over a message” and is considered strong. Thus, RSA’s signature scheme can authenticate transactions or identity in blockchain-like systems (though in practice many use ECC for key-size efficiency).
- Overall, RSA’s ability to provide both encryption and signature functionality makes it versatile: it secures web traffic (TLS/SSL), email and file encryption (PGP/GPG), and can be part of certificate authorities and secure messaging. These applications rely on RSA’s mathematical properties.

CHAPTER 7 – CONCLUSION

This project successfully implemented the RSA cryptosystem in Python, covering key generation, encryption/decryption (with and without OAEP padding), and digital signatures. The implementation was validated with sample messages and signatures. Performance measurements showed the expected behaviour: encryption (using the small public exponent) is fast, while decryption (using the large private exponent) is slower. RSA's correctness and security were confirmed by theory and practice. The report also surveyed RSA's applications in real systems (HTTPS, PGP, hybrid cryptography) and noted its limitations. In conclusion, RSA remains a fundamental building block of modern cryptography, and this implementation exercise deepened understanding of its mathematical and practical aspects.

CHAPTER 8 – FUTURE SCOPE

Quantum Computing Threat

Quantum computers running **Shor's Algorithm** can factor large numbers exponentially faster, making RSA insecure.

Post-Quantum Cryptography

Future systems will adopt **quantum-resistant algorithms** like:

- Lattice-based encryption (CRYSTALS-Kyber)
- Hash-based signatures (SPHINCS+)
- Code-based schemes (McEliece)

Transitioning from RSA to post-quantum cryptography is essential for long-term data security.

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